

PROFILE

MSc student of Geoinformatics and Cartography at the Faculty of Science, Palacký University Olomouc. Research interests: object detection from imagery and lidar with downstream spatial processing, evaluation of maps generated by large language models, and urban green space accessibility assessment. In parallel I work as a GIS and computer vision analyst in the CEDA Maps R&D team.

EDUCATION

MSc — Geoinformatics and Cartography (in progress) 2025 – 2027

Palacký University Olomouc · Faculty of Science, Dept. of Geoinformatics

- Focus: advanced spatial data processing, web cartography, remote sensing and image processing
- Master's thesis (in progress): *Evaluating LLM-generated maps against a traditional GIS workflow*
- Drones, laser scanning, 3D printing and virtual reality

BSc — Geoinformatics and Cartography 2022 – 2025

Palacký University Olomouc · Faculty of Science, Dept. of Geoinformatics

- Bachelor's thesis: *A user GIS toolbox for assessing the accessibility of parks and green spaces for urban residents* — grade B
- GIS, remote sensing, GNSS, geostatistics, computer and thematic cartography, spatial analysis and land-use planning

Maturita — Programming and Application Development / Internet of Things 2018 – 2022

Secondary School of Information Technology · Frýdek-Místek, Czechia

- Programming, web technologies, computer networks, databases and the fundamentals of electronics and IoT

PUBLICATIONS

- 2026 *Vehicle detection on large-area orthophotos*
Conference paper · PO GEO 2026 · github.com/MetrPikeska/vehicle-detection-ortophoto
- 2026 *Air pollution in the Moravian-Silesian Region*
Proceedings chapter · SYKAR 2026 · petrmikeska.cz/sykar2026_kapitola_Mikeska.pdf
- 2025 *A user GIS toolbox for assessing the accessibility of parks and green spaces for urban residents*
Bachelor's thesis · Dept. of Geoinformatics, Palacký University Olomouc · geoinformatics.upol.cz/dprace/bakalarske/mikeska25
- in progress *Evaluating LLM-generated maps against a traditional GIS workflow*
Master's thesis · Dept. of Geoinformatics, Palacký University Olomouc

RESEARCH PROJECTS

Field Data Collection Coordinator & Data Analyst — Palacký University Olomouc 2026 – present

Olomouc · Dept. of Geoinformatics, Faculty of Arts and Science and Technology Park, with Advolution · Visual attention and perception research in urban environments using eye-tracking

- Coordinating 40–60 field sessions recording participants' visual attention with eye-tracking glasses — logistics, route timing and participant communication
- Preprocessing recordings in Tobii Pro Lab: event creation, segmentation and ROI definition for evaluating visual exposure to outdoor advertising
- Collecting and managing spatial field data (GPS/GPX tracks) and synchronising it with gaze recordings; preparing data outputs for analysis
- Interdisciplinary collaboration across geoinformatics, psychology, academia and the private sector

3D Modelling Specialist — Palacký University Olomouc December 2025 – February 2026

Olomouc, Czechia · Modelling, processing and production of large-scale 3D models of two grottos including lighting and water installation

- 3D modelling of objects for physical installation and 3D printing; adjusting and optimising geometry for manufacturing (Blender, 3D Builder)

EXPERIENCE

GIS & Computer Vision Analyst (contractor) — CEDA Maps June 2026 – present

Brno, remote

- Building a GeoAI pipeline for automatic detection and classification of road signs from mobile mapping — from data annotation through model training to production deployment
- Localising signs without precise GNSS: multi-view triangulation of observations from dozens of passes, least-squares refinement and fusion into a single entity; monocular distance estimation (Depth Anything V2, MoGe-2) as an independent source, validated by a null test and scenario synthetic data
- Calibration of a panoramic camera rig via COLMAP SfM — extrinsics and intrinsics for lifting detections into real-world space
- Detecting sign panels directly in the LiDAR point cloud (RIEGL) and cross-validating them against image detections
- Dataset engineering: per-track splits against data leakage, deduplication of sequential frames and class-imbalance control over tens of thousands of crops; synthetic scene augmentation for weather and lighting
- Clustering and deduplication of detections across multiple passes (DBSCAN, ST_ClusterDBSCAN) and spatial analysis of routes over StreetNet data in PostGIS
- Python, PyTorch, Ultralytics YOLOv8/OBB, SAHI, ONNX Runtime, OpenCV, Open3D, laspy, COLMAP, CVAT, PostGIS, GeoPandas, Shapely

Co-founder and technical lead — VečerkaPlus

April 2026 – present

Frýdek-Místek, Czechia · vecerkaplus.cz

- Co-owner and technical lead of a late-night drinks and snacks delivery service in Frýdek-Místek — the entire web platform built from scratch on React, Supabase and Vercel
- Delivery zone system built on spatial analysis: 1,093 grid points via the Google Distance Matrix API, so price and availability follow real driving distance rather than straight-line distance
- GIS market-opportunity analysis over RÚIAN, the 2021 Czech Census and OSM POIs — 265,783 inhabitants across a 630 km² zone
- Supabase backend: PostgreSQL with PL/pgSQL triggers and RLS policies, Vercel Edge Functions, automated notifications via Resend, a Telegram bot and Twilio
- Alongside development I run operations — purchasing, courier logistics and growth strategy; average revenue of CZK 452 per order at a 36.5 % margin

Web Developer & Administrator — Palacký University Olomouc

May 2023 – present

Olomouc, Czechia

- Running olomouckymajales.cz and meetup.upol.cz — 199,800 visitors and 452,000 page views in total
- Requested changes delivered within 24 hours and uninterrupted operation during key events; design and functional changes deployed to production with no downtime
- WordPress, YOTheme Builder

GIS & Remote Sensing Analyst — Skymaps s.r.o.

February 2025 – September 2025

Czechia

- Processing multispectral satellite imagery into thematic maps of soil potential
- Analysis and interpretation of remote sensing data: Sentinel in SNAP, GDAL, EO Browser; Python automation from batch image processing to final map outputs
- QGIS, ArcGIS Pro, SNAP, EO Browser, Python

Drone Operator — Aerial imaging (freelance)

2024 – 2025

Ostrava, remote

- Commissioned drone footage for corporate marketing materials; planning and flying missions in compliance with EASA A1/A3 rules (DJI Mini 3 Pro)

Web Developer — Weiron Dynamics

2024 (4 months)

Czechia

- Building corporate websites in WordPress

SELECTED PROJECTS

Vehicle Detection on Orthophotos

2025 – 2026

github.com/MetrPikeska/vehicle-detection-orthophoto

Vehicle detection on large-area orthophotos of Olomouc, fully integrated into a GIS workflow. YOLOv8-OBb with SAHI enables inference on images larger than the GPU can hold; DBSCAN clustering of detections and Voronoi-based address assignment in GeoPandas follow. An end-to-end pipeline from CVAT annotation through training and georeferencing to spatial analysis. Presented at the international POGEO 2026 conference.

Python · YOLOv8-OBb · SAHI · GeoPandas · Shapely · CUDA

VečerkaPlus

2026

vecerkaplus.cz

A production ordering platform for late-night delivery. React 19, Vite and TypeScript on the front end with a bespoke design system and no third-party UI framework; a Supabase backend with PostgreSQL, PL/pgSQL triggers and RLS. The delivery zone is computed from 1,093 grid points via the Google Distance Matrix API.

React · TypeScript · Supabase · PostgreSQL · Google Maps API · Vercel

GEOTE Klima

2025

petrmikeska.cz/geote

A full-stack Web GIS platform for visualising and querying climate and environmental data. PostGIS with spatial indexes behind a Flask REST API, and a Leaflet front end with dynamic filtering directly on the map.

PostGIS · PostgreSQL · Flask · REST API · Leaflet · JavaScript

Park Accessibility Toolbox

2025

github.com/MetrPikeska/park-accessibility-toolbox

An ArcGIS Python toolbox for assessing pedestrian accessibility of parks and green spaces, applied to Olomouc, Brno and Ostrava in a single workflow. Network analysis via Network Analyst, automated from data input to final maps with no manual steps. Bachelor's thesis, grade B.

ArcGIS Pro · ArcPy · Network Analyst · Python

Electoral Atlas — GWR analysis

2026

petrmikeska.cz/pogeo

Spatial analysis of the Czech Pirate Party's electoral success across 6,157 municipalities using geographically weighted regression. The GWR model (R^2 0.445) against OLS (R^2 0.188) removed spatial autocorrelation in the residuals; predictors from the 2021 Czech Census. The output is an interactive electoral atlas.

R · GWR · PostGIS · FastAPI · Leaflet · ArcGIS Pro

Roundabout Exit Detection

2025

github.com/MetrPikeska/01_Detekce_vyjezdu

A system for automatically counting vehicles leaving a roundabout. YOLOv8 detection with geometric tracking via Shapely polygons keeps IDs stable through occlusion; CUDA acceleration handles video sequences in real time even on a GTX 1060.

Python · YOLOv8 · OpenCV · Shapely · CUDA

github.com/MetrPikeska/ai-generated-map-evaluation

Master's thesis: a systematic comparison of maps generated by five large language models (ChatGPT, Claude, Gemini, Mistral, GitHub Copilot) against a traditional GIS workflow, using object-based metrics and cartographic analysis. A Python batch pipeline, with results in an interactive Leaflet map.

Python · GeoPandas · Leaflet · GIS metrics

VY GEO Opálená — real-time IoT Web GIS

2025

An interactive ski resort map with 3D terrain visualisation from a DEM, real-time weather and automatic skier counting from a camera stream. Leaflet and Three.js on the front end, FastAPI with WebSockets for live data.

Leaflet · Three.js · FastAPI · WebSocket · PHP/MySQL · YOLO ONNX

CONFERENCES AND PROFESSIONAL EVENTS

- 2026 POGEO 2026 — international conference, presented a paper
- 2026 ISSonVIS 2026 — International Spring School on Visualization, Dept. of Geoinformatics, Palacký University Olomouc; theme *Emotions and Map: The Affective Dimension of Visualisation*
- 2026 GeoArts — talk, Dept. of Geoinformatics, Palacký University Olomouc
- 2026 InstallFest 2026, Faculty of Electrical Engineering, CTU Prague

SKILLS

Languages & data-bases	Python · SQL / PL-pgSQL · JavaScript / TypeScript · R · Bash · C# · PostgreSQL / PostGIS · MySQL · Supabase
Computer vision & machine learning	PyTorch · Ultralytics YOLOv8 / OBB · SAHI · OpenCV · torchvision · Hugging Face Transformers · Depth Anything V2 · MoGe-2 · scikit-learn · ONNX Runtime · CVAT · CUDA
GIS & spatial data	QGIS · ArcGIS Pro / ArcPy · GDAL/OGR · GeoServer · MapServer · SNAP · GeoPandas · Shapely · pyproj · Rasterio · H3 · Folium · OGC standards · EPSG 5514 / 3045 / 4326
LiDAR, 3D & photogrammetry	Point cloud processing (RIEGL) · Open3D · laspy · CloudCompare · Agisoft Metashape · COLMAP (SfM) · multi-view triangulation · camera rig calibration · DTM/DSM · Blender · MeshLab
Web & backend	React · Vite · Leaflet · FastAPI · Flask · Node.js / Express · REST API · WebSocket · HTML/CSS · WordPress · Vercel
Tooling & operations	Git / GitHub · Linux / Ubuntu / WSL · Docker · pytest · NumPy · Pandas · SciPy · Matplotlib · Claude Code · GitHub Copilot
Domain & standards	TP 65 (Czech road sign standard) · HD maps · mobile mapping systems · GNSS and NMEA · TomTom / StreetNet · RÚIAN · Sentinel / remote sensing · INSPIRE

CERTIFICATIONS

October 2024	Remote pilot certificate, category A1/A3 · Civil Aviation Authority of the Czech Republic / EASA
May 2023	Python for Everyone · Esri
April 2026	ISSonVIS 2026 — International Spring School on Visualization · Dept. of Geoinformatics, Palacký University Olomouc
April 2020	Introduction to IoT · Cisco Networking Academy
December 2021	Snowboard instructor, APUL/D licence · Association of Professional Ski Instructors (APUL), valid to 2027

LANGUAGES

Czech (native) · English (B2)